



**MEREWETHER HIGH
SCHOOL**

Name: _____

Class: _____

Teacher: _____

2023

Year 10

Assessment Task 4

Mathematics

General

- Total working time – 85 minutes

Instructions

- Write using black or blue pen
- Calculators allowed
- Show ALL relevant mathematical reasoning and/or calculations
- Marks may not be awarded for poorly arranged work

Total marks:

- Attempt all questions

110

- Answer in the spaces provided

Section	Topic	Marks
I	Polynomials	/13
II	Logarithms	/26
III	Properties of geometrical figures	/24
IV	Graphing Parabolas	/22
V	Non-linear Relationships and Graphs	/25
	Total	/110

Section I Polynomials

Part A – Multiple choice (3 marks)

Circle the most correct answer.

1. Given $P(x) = 2x^3 - x - 2$, evaluate $P(-1)$

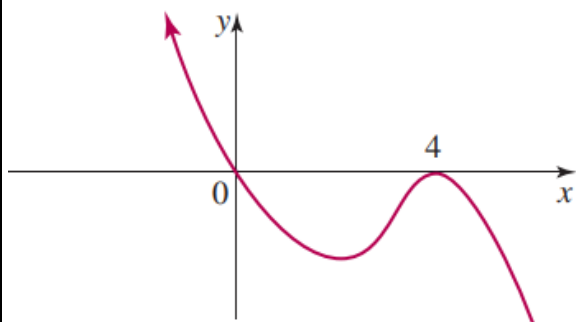
- A. -3
- B. -5
- C. 1
- D. -1

2. What is the degree and constant term of the polynomial :

$$P(x) = (2x^2 - 3)(x^2 - x - 1)$$

- A. 4 and -1
- B. 4 and 3
- C. 2 and 3
- D. 2 and 1

3. A possible equation for the polynomial shown below is best described by?



- A. $y = x(x+4)^2$
- B. $y = x(x-4)^2$
- C. $y = -x(x-4)$
- D. $y = -x(x-4)^2$

Part B – Short answer (10 marks)

1. What is the remainder when $P(x) = 3x^4 - 2x^2 - 3x - 5$ is divided by $A(x) = x - 1$?

[2]

.....

.....

.....

.....

.....

.....

.....

2.

a) Show that $x - 1$ is a factor of $P(x) = x^3 - 2x^2 - 11x + 12$

using the factor theorem

[1]

.....

.....

.....

.....

b) Perform the polynomial division shown:

[3]

$$x - 1 \overline{) x^3 - 2x^2 - 11x + 12}$$

- c) Write $P(x) = x^3 - 2x^2 - 11x + 12$ as a product of its linear factors **[1]**

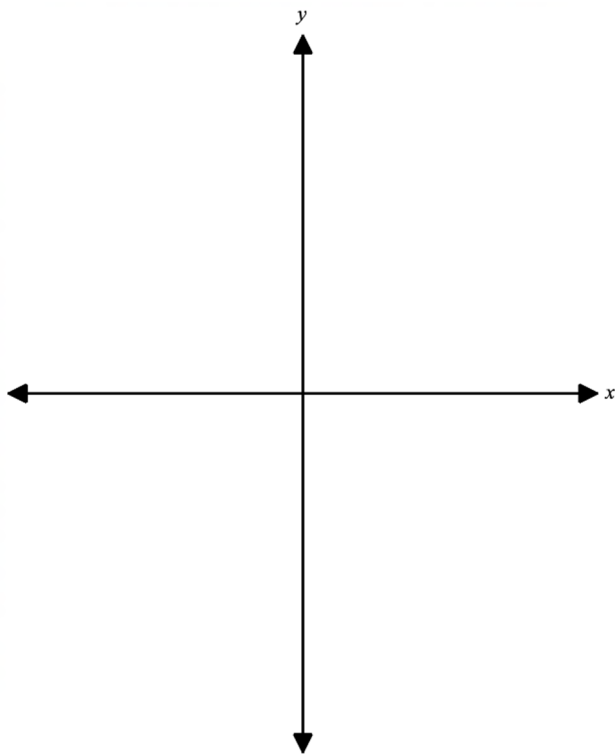
.....

.....

.....

.....

- d) Sketch the polynomial : **[3]**
 $P(x) = x^3 - 2x^2 - 11x + 12$ Showing all roots and the y -axis intercept.



BLANK WORKING SPACE

Section II Logarithms

Part A – Multiple choice (3 marks)

Circle the most correct answer.

1. Solve for y : $\log_y 4 = 2$

A. 4

B. 8

C. 2

D. 1

2. Simplify $3\log b + \log b^5$

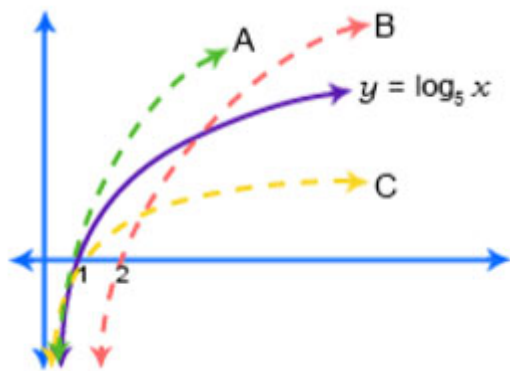
A. $3\log b + 5\log b^5$

B. $\log b^3 b^5$

C. $8\log b$

D. $8\log(b^8)$

3. The Graph of $y = \log_5 x$ is shown below. Which graph represents $y = \log_2 x$?



A. A

B. B

C. C

D. None of the above

Part B – Short answer (23 marks)

1. Solve for x :

(a) $\log_x 16 = 2$

[1]

.....

.....

.....

.....

.....

(b) $\log_5 x = 2$

[1]

.....

.....

(c) $2.1^x = 6$ (answer to 3 decimal places)

[2]

.....

.....

.....

.....

.....

.....

(d) $\log_b(2x-3) - \log_b 3 = \log_b 11$

[3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

2. Solve x using log base 10 to 3 decimal places

$$3000 \times 0.55^x = 500 \quad [3]$$

.....

.....

.....

.....

.....

.....

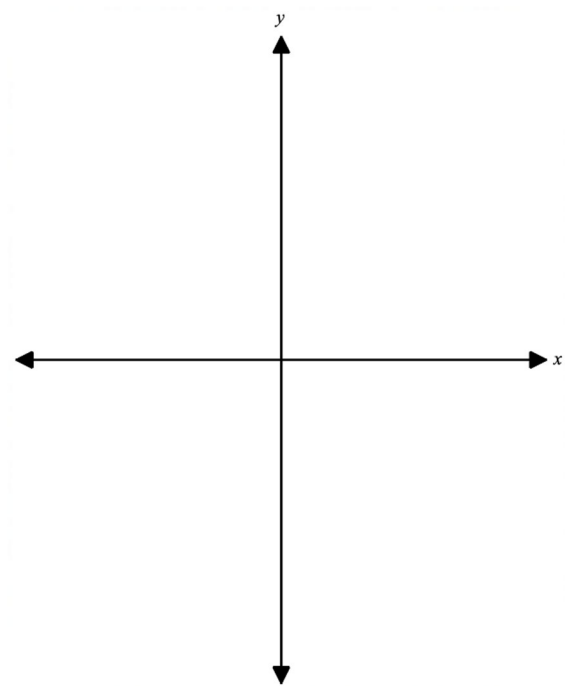
3. Write $3^3 = 27$ in log form. [1]

.....

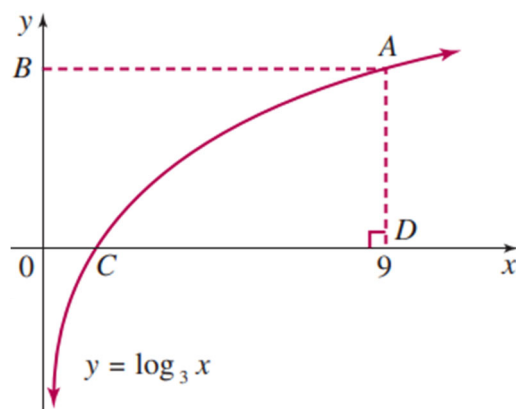
.....

.....

4. Sketch the graph of $y = \log(x + 2)$, showing all important features [3]



5.



Find the co-ordinates A, D, C. [3]

.....

.....

.....

.....

.....

.....

6. If $\log_x 2 = 0.123$ and $\log_x 3 = 0.234$ find:

(a) $\log_x 8$ [2]

.....

.....

.....

.....

(b) $\log_x (1.5)$ [2]

.....

.....

.....

(c) $\log_x (6x)$ [2]

.....

.....

.....

Section III Properties of Geometrical Figures

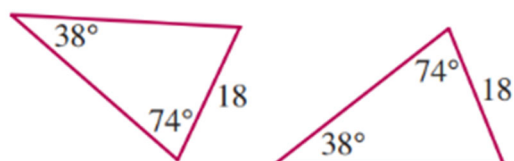
Part A – Multiple Choice (2 marks)

Select the correct answer.

1. Which quadrilateral has two pairs of equal adjacent sides with diagonals that intersect at right angles?

- A. Rectangle
- B. Kite
- C. Trapezium
- D. Parallelogram

2. State the test that would be used to prove that the pair of triangles is congruent



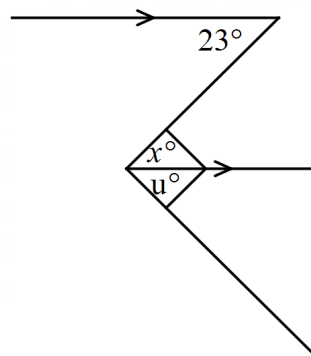
- A. SAS
- B. AAS
- C. SSS
- D. RHS

Part B – Short Answer (22 marks)

Answer all questions in the spaces provided.

1. Find the value of the pronumerals in the following giving reasons.

a) [2]



.....

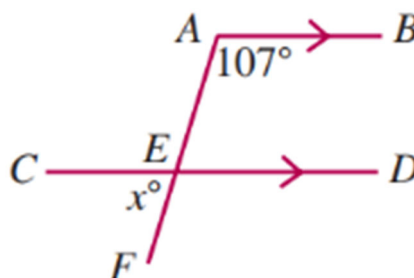
.....

.....

.....

.....

b) [2]



.....

.....

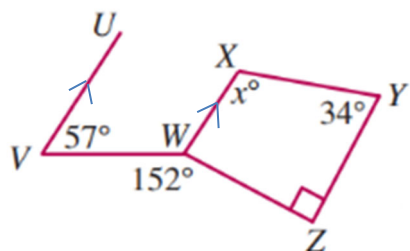
.....

.....

.....

c)

[3]



.....

.....

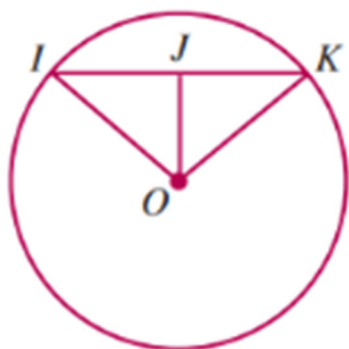
.....

.....

.....

2. O is the centre of the circle and $OJ \perp IK$.
Prove that OJ bisects $\angle IOK$.

[4]



.....

.....

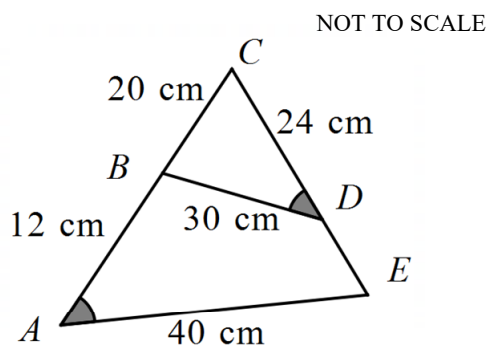
.....

.....

.....

3. In the diagram $AB = 12\text{cm}$, $BC = 20\text{cm}$
 $CD = 24\text{cm}$, $BD = 30\text{cm}$ and $AE = 40\text{cm}$

a) Prove triangles CBD and ACE are similar. [3]



.....

.....

.....

.....

.....

.....

.....

.....

.....

b) Find the length of DE
Round to nearest cm

[1]

.....

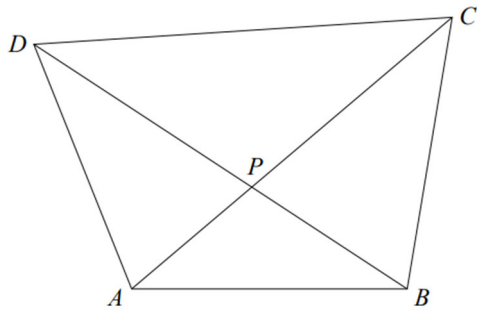
.....

.....

.....

.....

4. $ABCD$ is a quadrilateral. The diagonals AC and BD intersect at P .



$$AD=BC \text{ and } AC=BD.$$

- (i) Show that triangles ABC and ABD are congruent.[3]

.....

.....

.....

.....

- (ii) Show that triangle ABP is isosceles. [2]

.....

.....

.....

.....

.....

- (iii) Hence show that triangle CDP is isosceles. [2]

.....

.....

.....

.....

.....

.....

Section IV Graphing Parabolas

Part A – Multiple choice (3 marks)

Circle the most correct answer.

1. How many roots has the quadratic,

$$y = 3x^2 - 8x + 12.$$

- A. 0
- B. 1
- C. 2
- D. 3

2. What is the vertex of $y = (x+5)^2 + 1$

- A. (5,1)
- B. (-5,-1)
- C. (-5,1)
- D. (5,-1)

3. If $y = x^2$ is reflected in the x axis and then moved up 2 units the equation would be:

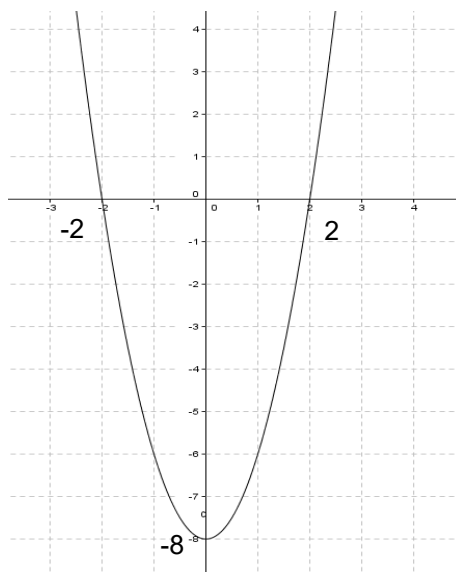
- A. $y = x^2 - 2$
- B. $y = -x^2 + 2$
- C. $y = -x^2 - 2$
- D. $y = -(x-2)^2$

Part B – Short answer (19 marks)

1. The parabola shown had the equation of the form $y = Ax^2 + C$.

Find the equation of the parabola .

[3]



.....

.....

.....

.....

.....

.....

.....

.....

2. The equations below are known to intersect, at $x = -1$ and one other value. Find the other x value of intersection. (Simultaneous equations)

$$y = x^2 + x + 1 \text{ and } y = 4x + 5.$$

[2]

.....

.....

.....

.....

3. Sketch the graph of $y = x^2 - 2x - 8$

a) Find the x -intercepts [1]

.....

.....

.....

.....

b) Find the y -intercept. [1]

.....

.....

c) Find the axis of symmetry. [1]

.....

.....

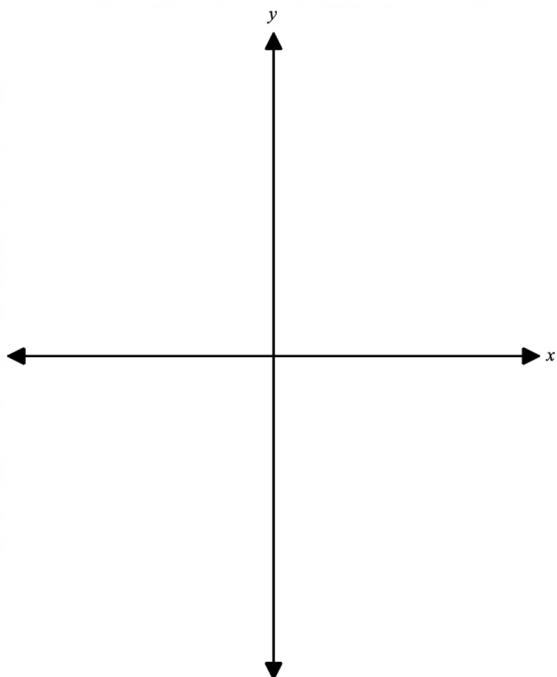
.....

d) Find the vertex. [2]

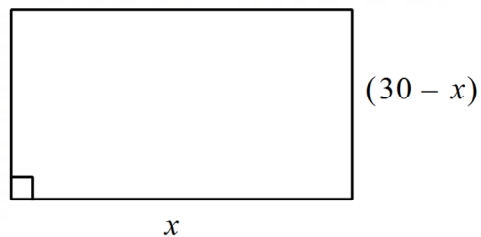
.....

.....

e) Hence sketch the graph of $y = x^2 - 4x - 8$ showing all important features. [2]



4. A piece of wire 60cm long is bent in the shape of a rectangle. Find the maximum area of the rectangle [3]



.....

.....

.....

.....

.....

.....

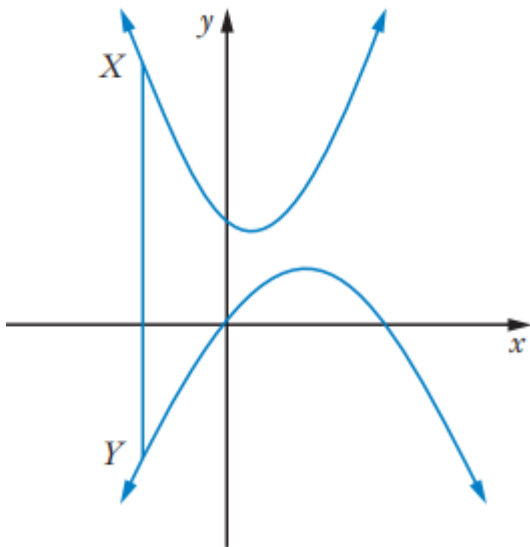
.....

.....

.....

.....

5. X is a point on $y = x^2 - 2x + 5$.
 Point Y lies directly below X and is on the
 curve $y = 4x - x^2$.



- a) Show that the distance, between X and Y is
 given by $d = 2x^2 - 6x + 5$.

[2]

.....

.....

.....

.....

.....

.....

.....

- b) X and Y move together on their respective
 parabolas, what is the minimum distance
 they are apart?

[2]

.....

.....

.....

.....

BLANK WORKING SPACE

Section V Non-linear Relationships and Graphs

Part A – Multiple choice (2 marks)

Circle the most correct answer.

1. Which of the following graph(s) represent a function?

- A. $x = y^2$
- B. $x^2 + y^2 = 1$
- C. $y = 2^{-x} + 1$
- D. $y^2 = 1 - x^2$

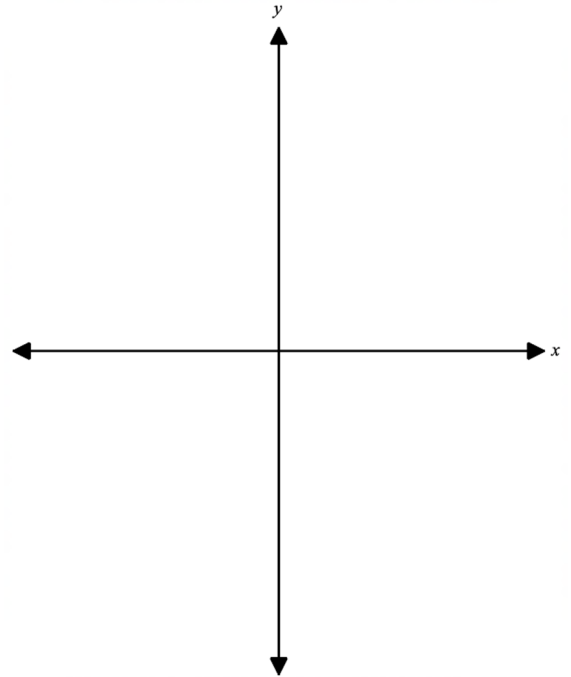
2. What is the centre and radius of the circle $(x)^2 - 25 = -(y + 3)^2$?

- A. $(0, -3), r = 25$
- B. $(0, 3), r = 5$
- C. $(0, -3), r = 5$
- D. $(5, 3), r = 5$

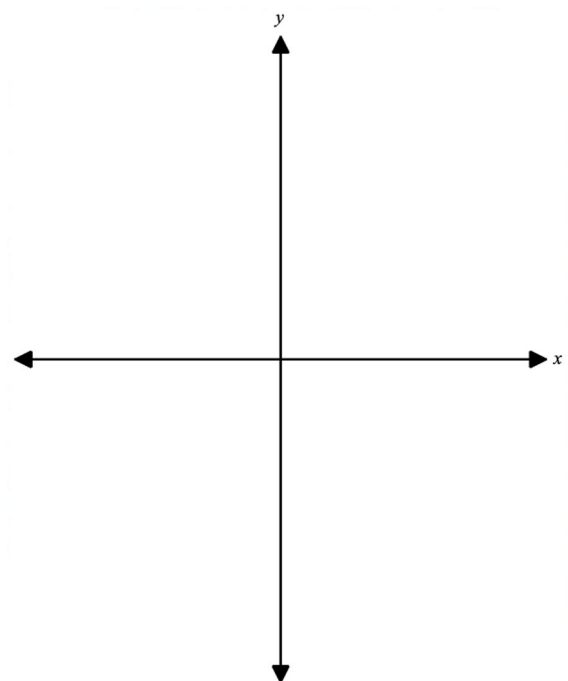
Part B – Short answer (23 marks)

1. Graph the following functions showing all key features .

a) $y = 3^{-x} + 1$ [3]

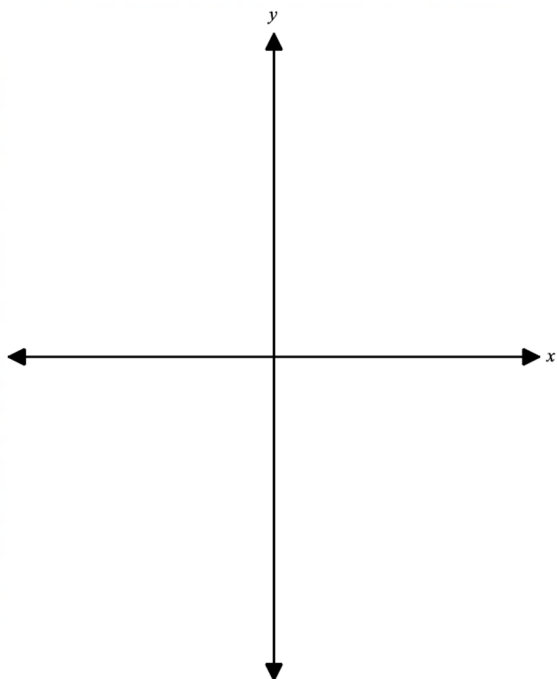


b) $y = (x - 1)^3 - 1$ [3]



c) $-xy = -4$

[3]



Domain:.....

Range:

2. Find the inverse function of $y = 2 - \frac{x}{3}$.

[2]

.....

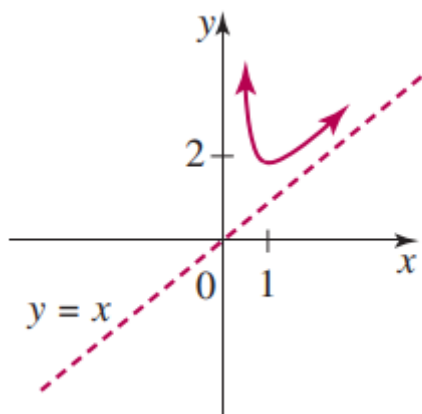
.....

.....

.....

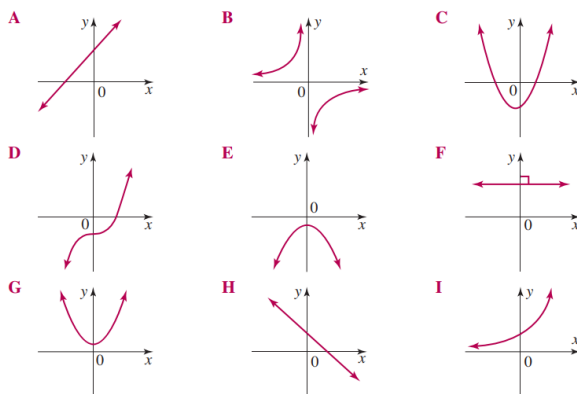
.....

3. Sketch the inverse function for the graph below [1]



4. Match the appropriate equation to each graph shown below:

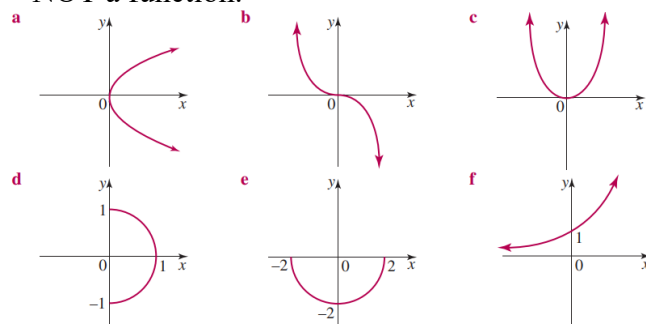
[9]



Function	Corresponding letter
$y = 4 - x$	
$y = (2.5)^x$	
$3 = x^2 - y$	
$y = x + 4$	
$y + x^2 + 3 = 0$	
$y = x^3 - 2$	
$y - x^2 = 0.3$	
$y = 2$	
$y = \frac{-3}{x}$	

5. Circle any of the graphs below that are NOT a function.

[2]



End of Task

[illegible]



2023

Solutions

Year 10

Assessment Task 4

Mathematics

General

- Total working time – 85 minutes

Instructions

- Write using black or blue pen
- Calculators allowed
- Show ALL relevant mathematical reasoning and/or calculations
- Marks may not be awarded for poorly arranged work

Total marks:

110

- Attempt all questions
- Answer in the spaces provided

Section	Topic	Marks
I	Polynomials	/13
II	Logarithms	/26
III	Properties of geometrical figures	/24
IV	Graphing Parabolas	/22
V	Non-linear Relationships and Graphs	/25
	Total	/110

Section I Polynomials

Part A – Multiple choice (3 marks)

Circle the most correct answer.

1. Given $P(x) = 2x^3 - x - 2$, evaluate $P(-1)$

☒ A. -3

B. -5

C. 1

D. -1

2. What is the degree and constant term of the polynomial :

$$P(x) = (2x^2 - 3)(x^2 - x - 1)$$

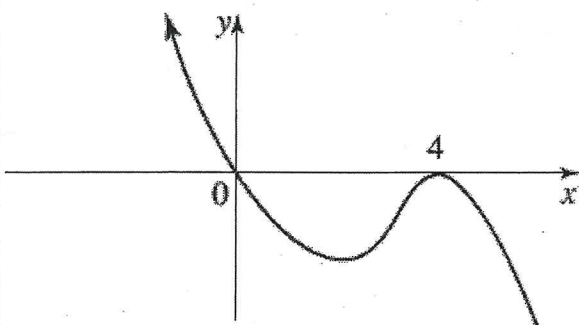
A. 4 and -1

☒ B. 4 and 3

C. 2 and 3

D. 2 and 1

3. A possible equation for the polynomial shown below is best described by?



A. $y = x(x+4)^2$

B. $y = x(x-4)^2$

C. $y = -x(x-4)$

☒ D. $y = -x(x-4)^2$

Part B – Short answer (10 marks)

1. What is the remainder when

$P(x) = 3x^4 - 2x^2 - 3x - 5$ is divided by

$A(x) = x - 1$?

[2]

$$\begin{aligned} P(1) &= 3(1)^4 - 2(1)^2 - 3(1) - 5 \\ &= 3 - 2 - 3 - 5 \\ &= -7 \end{aligned}$$

2.

a) Show that $x - 1$ is a factor of $P(x) = x^3 - 2x^2 - 11x + 12$

using the factor theorem

[1]

$$\begin{aligned} P(1) &= 1^3 - 2(1)^2 - 11(1) + 12 \\ &= 1 - 2 - 11 + 12 \\ &= 0 \end{aligned}$$

b) Perform the polynomial division shown:

[3]

$$\begin{array}{r} x^2 - x - 12 \\ x-1 \overline{) x^3 - 2x^2 - 11x + 12} \\ \underline{x^3 - x^2} \\ -x^2 - 11x \\ \underline{-x^2 + x} \\ -12x + 12 \\ \underline{-12x + 12} \\ 0 \end{array}$$

- c) Write $P(x) = x^3 - 2x^2 - 11x + 12$ as a product of its linear factors [1]

$$P(x) = x^3 - 2x^2 - 11x + 12$$

$$= (x-1)(x^2 - x - 12)$$

$$= (x-1)(x-4)(x+3)$$

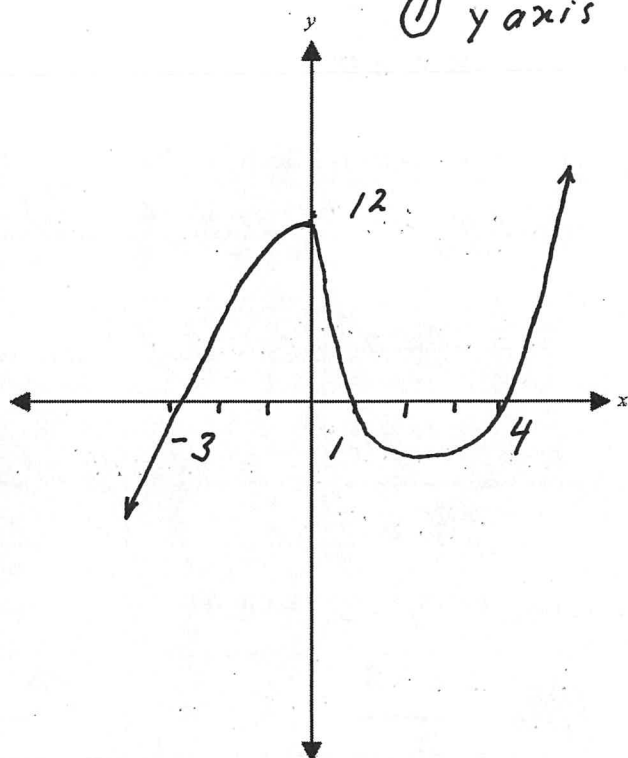
- d) Sketch the polynomial : [3]

$P(x) = x^3 - 2x^2 - 11x + 12$ Showing all roots and the y-axis intercept.

① roots

① shape

① y axis



Section II Logarithms

Part A – Multiple choice (3 marks)

Circle the most correct answer.

1. Solve for y : $\log_y 4 = 2$

A. 4

B. 8

☒ C. 2

D. 1

2. Simplify $3\log b + \log b^5$

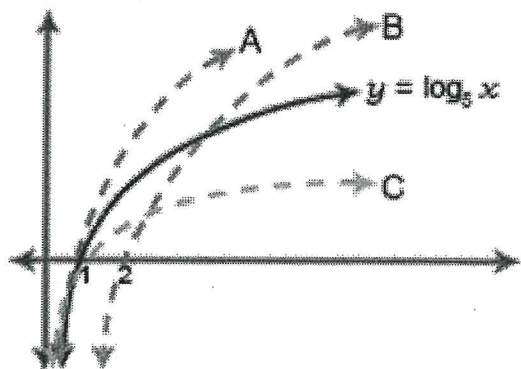
A. $3\log b + 5\log b^5$

B. $\log b^3 b^5$

☒ C. $8\log b$

D. $8\log(b^8)$

3. The Graph of $y = \log_5 x$ is shown below. Which graph represents $y = \log_2 x$?



☒ A. A

B. B

C. C

D. None of the above

Part B – Short answer (23 marks)

1. Solve for x :

(a) $\log_x 16 = 2$

[1]

$$x^2 = 16$$

$$x = 4$$

(b) $\log_5 x = 2$

[1]

$$5^2 = x$$

$$x = 25$$

(c) $2.1^x = 6$ (answer to 3 decimal places)

[2]

$$\log_{10} 2.1^x = \log_{10} 6 \quad (1)$$

$$x = \log_{10} 2.1^6 = \log_{10} 6$$

$$x = \frac{\log_{10} 6}{\log_{10} 2.1} = 2.415 \quad (1)$$

(d) $\log_b (2x-3) - \log_b 3 = \log_b 11$

[3]

$$\log_b \frac{2x-3}{3} = \log_b 11 \quad (1)$$

$$\therefore \frac{2x-3}{3} = 11 \quad (1)$$

$$2x - 3 = 33$$

$$2x = 36$$

$$x = 18 \quad (1)$$

2. Solve x using log base 10 to 3 decimal places

$$3000 \times 0.55^x = 500$$

[3]

$$0.55^x = \frac{500}{3000} \quad (1)$$

$$0.55^x = \frac{1}{6}$$

$$x \log_{10} 0.55 = \log_{10} \frac{1}{6} \quad (1)$$

$$x = \frac{\log_{10} \frac{1}{6}}{\log_{10} 0.55} = 2.997 \quad (1)$$

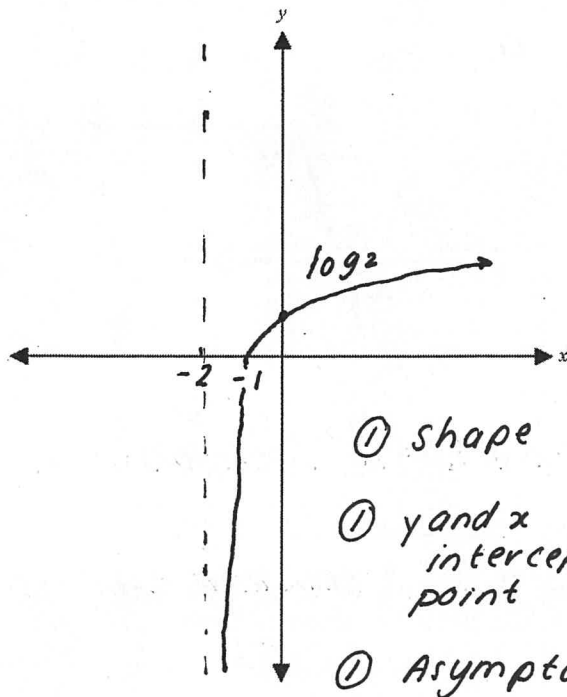
3. Write $3^3 = 27$ in log form.

[1]

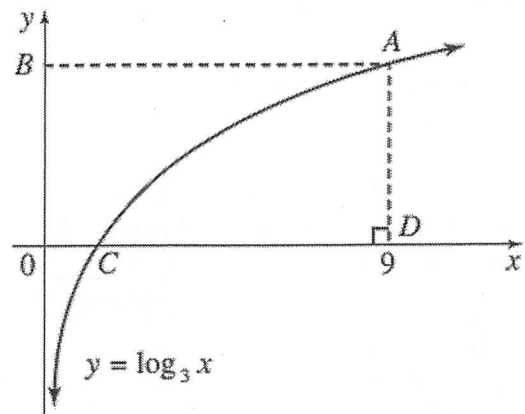
$$\log_3 27 = 3$$

4. Sketch the graph of $y = \log(x+2)$, showing all important features

[3]



5.



Find the co-ordinates A, D, C.

[3]

$$A(9, 2) \quad (1)$$

$$D(9, 0) \quad (1)$$

$$C(1, 0) \quad (1)$$

6. If $\log_x 2 = 0.123$ and $\log_x 3 = 0.234$ find:

[2]

$$(a) \log_x 8 = \log_x 2^3 \quad (1)$$

$$= 3 \log_x 2$$

$$= 3 \times 0.123$$

$$= 0.369 \quad (1)$$

$$(b) \log_x(1.5) = \log_x\left(\frac{3}{2}\right) \quad (1)$$

[2]

$$= \log_x 3 - \log_x 2$$

$$= 0.234 - 0.123$$

$$= 0.111 \quad (1)$$

$$(c) \log_x(6x)$$

[2]

$$= \log_x 6 + \log_x x$$

$$= \log_x 2 + \log_x 3 + 1$$

$$= 0.123 + 0.234 + 1$$

$$= 1.357$$

Section III Properties of Geometrical Figures

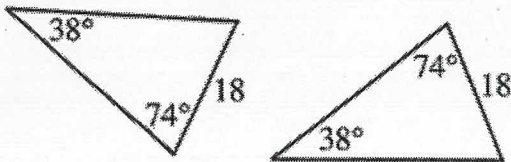
Part A – Multiple Choice (2 marks)

Select the correct answer.

1. Which quadrilateral has two pairs of equal adjacent sides with diagonals that intersect at right angles?

- A. Rectangle
☒ B. Kite
 C. Trapezium
 D. Parallelogram

2. State the test that would be used to prove that the pair of triangles is congruent



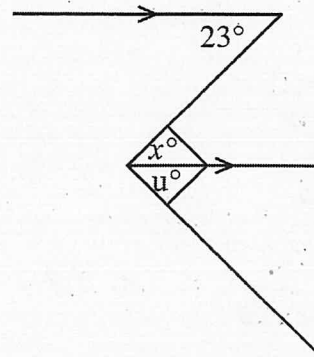
- A. SAS
☒ B. AAS
 C. SSS
 D. RHS

Part B – Short Answer (22 marks)

Answer all questions in the spaces provided.

1. Find the value of the pronumerals in the following giving reasons.

a) [2]

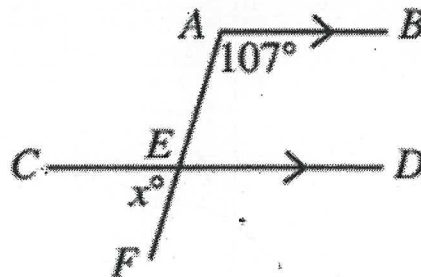


$$x^\circ = 23^\circ \text{ (alternate } \angle\text{s, } \textcircled{1} \text{ parallel lines)}$$

$$u^\circ = 90 - 23 \text{ (right angle)} \textcircled{1}$$

$$= 67^\circ$$

b) [2]



$$\angle AED = 73^\circ \text{ (cointerior } \angle\text{s, } AB \parallel ED) \textcircled{1}$$

$$x = 73^\circ \text{ (vertically opposite } \angle\text{s)} \textcircled{1}$$

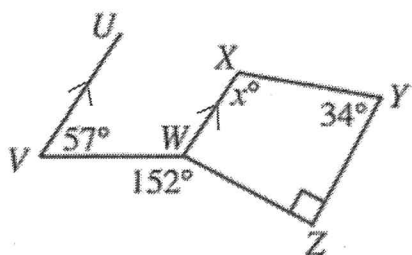
OR

$$\angle AEC = 107^\circ \text{ (alternate } \angle\text{s, } AB \parallel CD) \textcircled{1}$$

$$x = 180^\circ - 107^\circ \text{ (angles on straight line)} \textcircled{1}$$

c)

[3]



$$W = 180 - 57 = 123^\circ \text{ (co-interior } \angle\text{s, } UV \parallel WX \text{)}$$

$$\angle ZWX = 360 - (123 + 152) \text{ (}\angle\text{s at } WX \text{)}$$

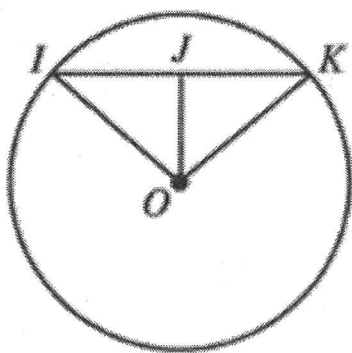
$$= 85^\circ \text{ (a point)}$$

$$x = 360 - (85 + 34 + 90) \text{ (}\angle\text{sum of quad)}$$

$$= \cancel{174^\circ} \quad 151^\circ$$

2. O is the centre of the circle and $OJ \perp IK$.
Prove that OJ bisects $\angle IOK$.

[4]



in $\triangle OJI$ and $\triangle OJK$,

$$OI = OK \text{ (equal radii)} \quad ①$$

$$OJ \text{ is common} \quad ①$$

$$\angle IJO = \angle KJO \text{ (} OJ \perp IK \text{)} \quad ①$$

$$\triangle OJI \equiv \triangle OJK \text{ (RHS)}$$

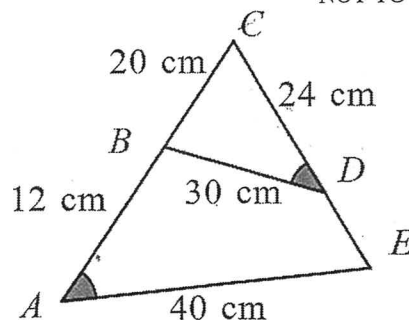
$$IJ = JK \text{ (sides in } \equiv \Delta \text{)}$$

$$\therefore OJ \text{ bisects } \angle IOK \quad ①$$

3. In the diagram $AB = 12\text{ cm}$, $BC = 20\text{ cm}$
 $CD = 24\text{ cm}$, $BD = 30\text{ cm}$ and $AE = 40\text{ cm}$

a) Prove triangles CBD and CAE are similar. [3]

NOT TO SCALE



in $\triangle CBD$ and $\triangle CAE$

$$\angle CBD = \angle CAE \text{ (given)} \quad ①$$

$$\frac{CD}{CA} = \frac{24}{32} = \frac{3}{4} \quad ①$$

$$\frac{BD}{AE} = \frac{30}{40} = \frac{3}{4} \quad ①$$

$\triangle CBD \sim \triangle CAE$ (two pairs of corresponding sides in proportion and included angle)

b) Find the length of DE
Round to nearest cm

[1]

$$\frac{CE}{20} = \frac{40}{30}$$

$$CE = \frac{40 \times 20}{30}$$

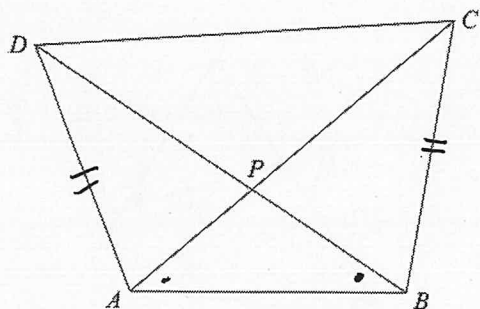
$$= 26.6$$

$$\therefore DE = 26.6 - 24$$

$$= 2.6$$

$$= 3 \text{ (rounded to nearest cm)}$$

4. $ABCD$ is a quadrilateral. The diagonals AC and BD intersect at P .



$$AD=BC \text{ and } AC=BD.$$

- (i) Show that triangles ABC and ABD are congruent. [3]

in $\triangle ABC$ and $\triangle ABD$

AB is common ①

$AD=BC$ (given) ①

$AC=BD$ (given) ①

$\therefore \triangle ABC \equiv \triangle BAD$ (SSS)

- (ii) Show that triangle ABP is isosceles. [2]

$\angle CAB = \angle DBA$ (corresponding $\angle$ s,
congruent Δ s) ①

$\therefore \triangle ABP$ is isos (base $\angle$ s
equal) ①

- (iii) Hence show that triangle CDP is isosceles. [2]

$AC=BD$ (corresponding sides
in congruent Δ) ①

$AP=PB$ (isos Δ)

$\therefore DP=CP$ ①

$\therefore \triangle CDP$ is isos (2 equal
sides)

Section IV Graphing Parabolas

Part A – Multiple choice (3 marks)

Circle the most correct answer.

1. How many roots has the quadratic,

$$y = 3x^2 - 8x + 12.$$

- A. 0
B. 1
C. 2
D. 3

2. What is the vertex of $y = (x+5)^2 + 1$

- A. (5,1)
B. (-5,-1)
C. (-5,1)
D. (5,-1)

3. If $y = x^2$ is reflected in the x axis and then moved up 2 units the equation would be:

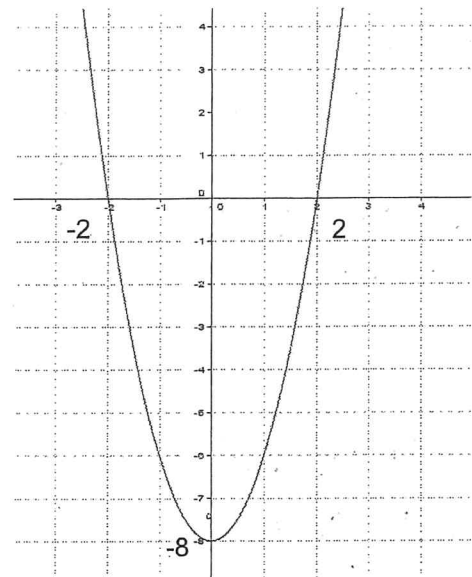
- A. $y = x^2 - 2$
B. $y = -x^2 + 2$
C. $y = -x^2 - 2$
D. $y = -(x-2)^2$

Part B – Short answer (19 marks)

1. The parabola shown had the equation of the form $y = Ax^2 + C$.

Find the equation of the parabola.

[3]



$$x=0, y=-8 \quad (0, -8)$$

$$\therefore -8 = A(0)^2 + C$$

$$C = -8$$

$$y = Ax^2 - 8$$

$$x=2 \quad y=0$$

$$0 = A(2)^2 - 8$$

$$8 = 4A$$

$$A = 2$$

$$\therefore y = 2x^2 - 8$$

2. The equations below are known to intersect, at $x = -1$ and one other value. Find the other x value of intersection. (Simultaneous equations)

$$y = x^2 + x + 1 \quad \text{and} \quad y = 4x + 5$$

$$x^2 + x + 1 = 4x + 5$$

$$x^2 - 3x - 4 = 0$$

$$(x-4)(x+1) = 0$$

$$x = 4, -1$$

$\therefore$ other pt. of intersection at $x = 4$

3. Sketch the graph of $y = x^2 - 2x - 8$

a) Find the x-intercepts [1]

$$(x-4)(x+2) = 0$$

$$x = 4, x = -2$$

b) Find the y-intercept. [1]

$$y = 0^2 - 2(0) - 8$$

$$(0, -8)$$

c) Find the axis of symmetry. [1]

$$x = -b/2a$$

$$= \frac{2}{2} = 1$$

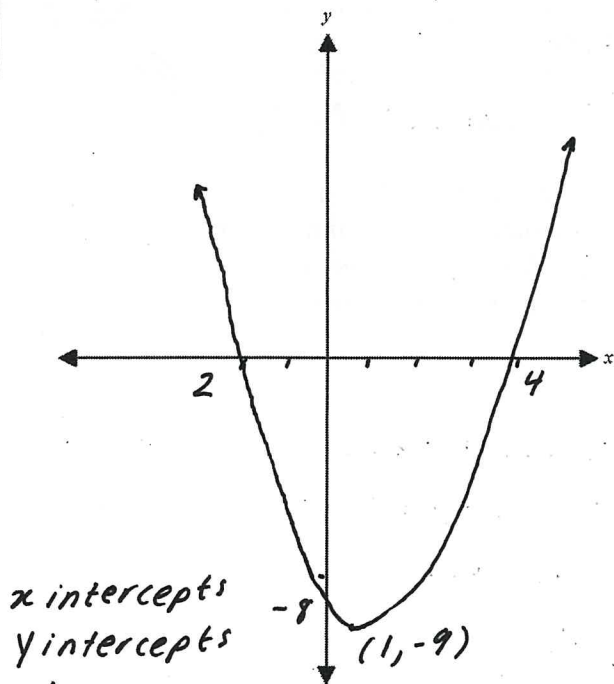
d) Find the vertex. [2]

$$\text{on } x=1, \text{ ①}$$

$$y = 1^2 - 2(1) - 8$$

$$= -9 \text{ ①}$$

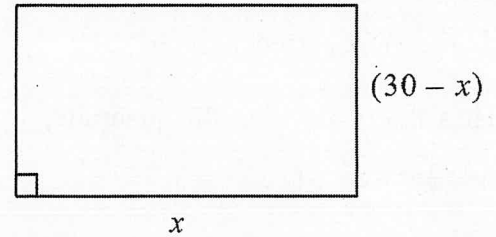
e) Hence sketch the graph of $y = x^2 - 4x - 8$ showing all important features. [2]



x intercepts
y intercepts
shape
vertex

① at least 2
② all of these

4. A piece of wire 60cm long is bent in the shape of a rectangle. Find the maximum area of the rectangle [3]



$$A = x(30-x)$$

$$= 30x - x^2$$

$$= -x^2 + 30x \text{ ①}$$

max on axis

$$x = \frac{-b}{2a}$$

$$= \frac{-30}{2(-1)} = \frac{30}{2}$$

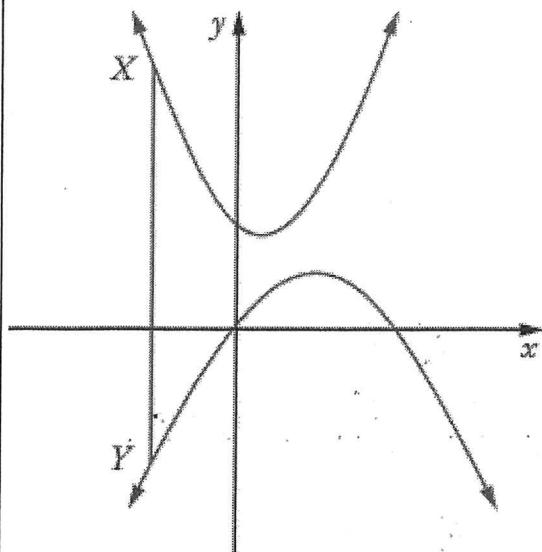
$$= 15 \text{ ①}$$

$$A = 15(30-15)$$

$$= 225 \text{ cm}^2 \text{ ①}$$

5. X is a point on $y = x^2 - 2x + 5$.

Point Y lies directly below X and is on the curve $y = 4x - x^2$.



- a) Show that the distance, between X and Y is given by $d = 2x^2 - 6x + 5$.

[2]

$$d = x^2 - 2x + 5 - (4x - x^2) \quad ①$$

$$= x^2 - 2x + 5 - 4x + x^2$$

$$= 2x^2 - 6x + 5 \quad ①$$

- b) X and Y move together on their respective parabolas, what is the minimum distance they are apart?

[2]

$$x = \frac{-(-6)}{2(2)} = \frac{3}{2} \quad ①$$

$$y = 2\left(\frac{3}{2}\right)^2 - 6\left(\frac{3}{2}\right) + 5$$

$$= 4.5 - 9 + 5$$

$$= 0.5 \quad ①$$

BLANK WORKING SPACE

Section V Non-linear Relationships and Graphs

Part A – Multiple choice (2 marks)

Circle the most correct answer.

1. Which of the following graph(s) represent a function?

A. $x = y^2$

B. $x^2 + y^2 = 1$

☒ C. $y = 2^{-x} + 1$

D. $y^2 = 1 - x^2$

2. What is the centre and radius of the circle $(x)^2 - 25 = -(y+3)^2$?

A. $(0, -3), r = 25$

B. $(0, 3), r = 5$

☒ C. $(0, -3), r = 5$

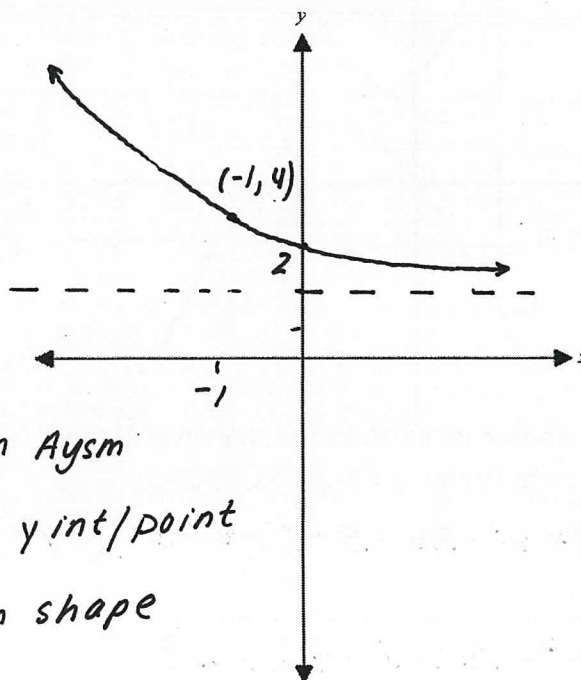
D. $(5, 3), r = 5$

Part B – Short answer (23 marks)

1. Graph the following functions showing all key features.

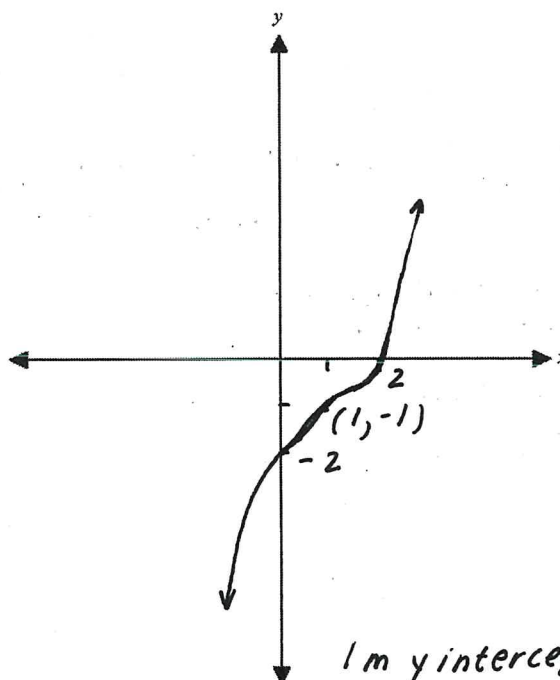
a) $y = 3^{-x} + 1$

[3]



b) $y = (x-1)^3 - 1$

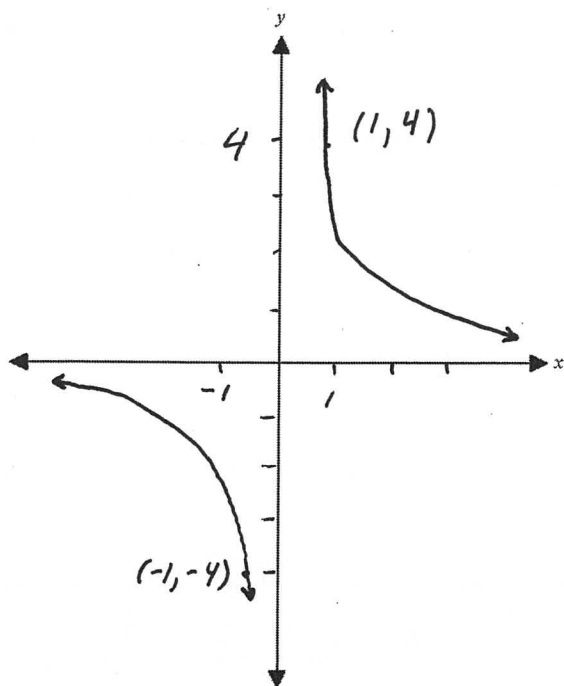
[3]



c) $-xy = -4$

[3]

1m shape
1m points x



Domain: ...all x ... $x \neq 0$...

Range: ...all y ... $y \neq 0$...

① correct
D and
R.

2. Find the inverse function of $y = 2 - \frac{x}{3}$.

[2]

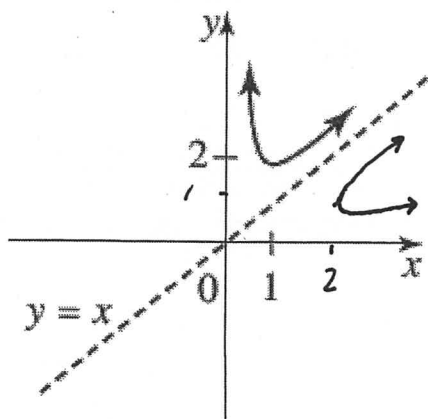
$$x = 2 - \frac{y}{3}$$

$$x - 2 = -\frac{y}{3}$$

$$3x - 6 = -y$$

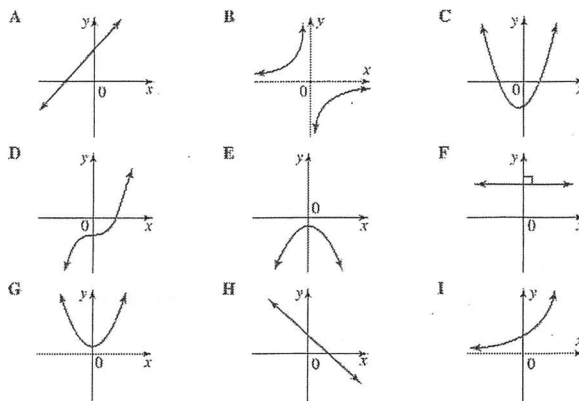
$$-3(x - 2) = y$$

3. Sketch the inverse function for the graph below [1]



4. Match the appropriate equation to each graph shown below:

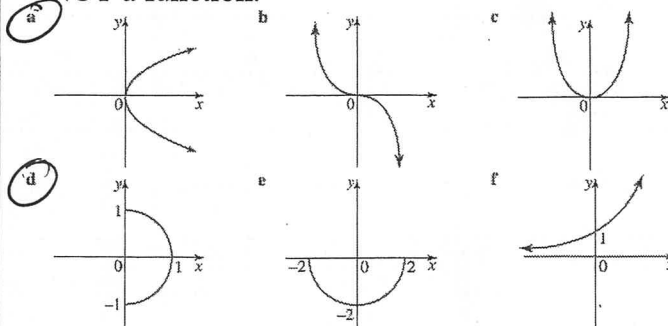
[9]



Function	Corresponding letter
$y = 4 - x$	H
$y = (2.5)^x$	I
$3 = x^2 - y$	C
$y = x + 4$	A
$y + x^2 + 3 = 0$	E
$y = x^3 - 2$	D
$y - x^2 = 0.3$	G
$y = 2$	F
$y = \frac{-3}{x}$	B

5. Circle any of the graphs below that are NOT a function.

[2]



① mark each.

End of Task

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the paper.